

NeuroWrite: A Smart Pen for Sensor-Based Handwriting Screening of Dysgraphia and ADHD

Abstract

Dysgraphia and Attention Deficit Hyperactivity Disorder (ADHD) are typically seen as problems with handwriting in young children. However, existing assessment methods rely on subjective observation, specialist assessment, and extended monitoring. The present research proposes NeuroWrite, a low-cost, sensor-based smart pen designed to allow objective and non-invasive early detection of handwriting-related motor abnormalities. The system utilizes two Force Sensitive Resistors (FSRs) to capture grip pressure changes and an MPU6050 accelerometer-gyroscope to record the acceleration cycle while writing. Data was collected from 60 individuals, composed of 20 control, 20 ADHD, and 20 dysgraphic samples, through pangram and shape-drawing exercises. The analysis identified specific motor patterns in each group demonstrating higher jerk and variability of motion in dysgraphic individuals, while ADHD participants revealed higher variability of pressure and less pressure-motion coordination. A web-based NeuroWrite dashboard has been developed to facilitate the feature extraction, visualization of data, and probability-based assessment of uploaded sensor data. Findings confirm the applicability of wearable sensors and data analysis in the early discovery of handwriting issues, with the NeuroWrite tool accessible for education specialists, doctors, and parents.

Keywords - NeuroWrite; Dysgraphia Screening; Attention Deficit Hyperactivity Disorder (ADHD); Smart Pen Technology; Handwriting Analysis; Neuromotor Assessment; Digital Biomarkers; Early Detection.

Introduction

The neurological learning disability known as dysgraphia has an impact on the ability to do handwriting, and is estimated to affect 5-20% of school aged children; dysgraphia leads to learning problems and low self esteem [1]. Dysgraphia is diagnosed by observing a child's handwriting; however, this subjective visual judgement takes several weeks and delays early intervention. Researchers understood that handwriting is complicated because it is a sensorimotor activity that involves both visual, tactile (touch) and proprioceptive (sense of body position) elements [2]. Problems with handwriting involve irregular motion when writing (kinematics) and inconsistent amount of force being applied to the writing implement when writing (pressure) along with uneven muscle control (motor control), which cannot all be captured using image based assessment alone [3]. Wearable sensing technologies, such as Inertial Measurement Units (IMUs) and Force Sensitive Resistors (FSRs), can provide an objective way to measure the kinematic and pressure (sensor) characteristics of children's handwriting by collecting data about the acceleration and angular velocity of movement from IMUs, and the amount of grip and writing force from pressure sensors. The purpose of this work is to propose and illustrate a multimodal (multiple levels of analysis), sensor-based (sensor driven) system for real-time assessment of dysgraphia. The results of this system will provide a quick, inexpensive, and objective method of assessing dysgraphia in children compared to traditional methods.

Literature Review

According to Chung et al. [1] dysgraphia is a writing disorder caused by a malfunction of the nervous system, resulting in a failure in coordination and recognition. This paper highlights the importance of finding objective early diagnostic tools in order to provide timely diagnosis. It is this idea that motivated the use of measurable signals indicating a person's ability to control motor behavior. Danna et al. [2] confirmed that handwriting is the result of multiple feedback and that keeping handwriting steady requires controlling force and movements with great accuracy. The results obtained by the researchers prove that FSR and IMU are suitable instruments for measuring neuromotor activity.

In a study conducted by Gargot et al., [3] the use of computer-assisted techniques demonstrated the contrast between the handwriting of dysgraphic and non-dysgraphic learners. The research showed that the handwriting of individuals with dysgraphia is characterized as inconsistent and unstable in terms of movement and control, which cannot be captured through static methods. This finding regarding the dynamic characteristics, such as velocity, consistency, and kinematics, corresponds with the results of the survey carried out by Rathod et al., [4] which presented alternative detection techniques.

In their study, Namitha et al. [5] demonstrate the shortcomings of image-based methods by developing a system that detects the presence of dysgraphia based only on visible features including stroke consistency, spacing, and letter formation. While offering effective accuracy, their approach ignores important process-related components like pressure and motion. It is precisely this limitation that drives researchers like Tao et al. [6] to develop sensor-driven approaches to identifying dysgraphia.

Wang et al., [7] used an IMU embedded in a pen and utilized for capturing ongoing three-dimensional movements which allows individual strokes and characters to be tracked instantly. Jeen-Shing Wang et al. [8] further support this form of handwriting recognition that is based on inertia sensors since it is proven that such systems can be superior to visual techniques. Bisio et al. [9] then determine what specific kinematic parameters—namely, speed, acceleration, and stroke smoothness—are of the greatest importance indicating that these measures convey the way motor skills influence the quality of handwriting and can serve as reliable indicators of handwriting behavior.

In order to promote reproducible research, Gupta et al. [10] have released a dataset containing IMU recordings of handwritten English letters, full of raw data about acceleration and angular velocity, so that feature extraction and model training can be accomplished without images. Building upon this data, Li et al. [11] proposed an advanced IMU handwriting recognition system that learns the necessary motion characteristics to ensure the recognition of handwriting of different users based on their individual styles and thus, solves scalability issues identified in previous studies. Mummadi et al. [12] proved that it is possible to utilize an IMU technology in real-time applications showing that an IMU glove can be deployed for gesture recognition which ensures on-device classification and low-latency feature extraction aimed at the development of advanced dysgraphia diagnostic systems based on either pen or glove.

According to Feder and colleagues [13], motor control, coordination, and cognitive abilities combine to produce the phenomenon of writing, noting that the poor development of fine motor abilities and improper integration of sensorimotor processes account for difficulties such as dysgraphia. Yang and colleagues [14] go on to provide essential information about wearable IMU technologies that need to be

solved, such as where to place their devices, data processing issues, noise, and drift. Finally, Drotar and colleagues [15] combine these pieces of knowledge to develop the rationale for using wearable IMUs in handwriting assessments, claiming that continuous and precise assessment of movements will improve diagnosis and treatment.

Even though great progress has been achieved in the area of digital handwriting analysis, many systems available on the market are based on the use of digitizing tablets, smart pens equipped with exclusive devices, or image-based handwriting assessment techniques. These methods are mainly described and researched by focus on the final written product rather than the biomechanical features the handwriting process involves. So far, few studies provide the high level of integration of the methodology of performing real-time pressure detection, motion analysis, data collection over the cloud and automated classification of the acquired data in one system. Therefore the proposed work presents a solution of the existing issues through using a combination of a smart pen integrated with an array of sensors along with a machine learning-based diagnostic dashboard aimed at the objective and easy handwriting evaluation.

Methodology

The workflow of the precise handwriting assessment system is illustrated in Figure 1. Here, the process starts from handwriting acquisition by means of the smart pen and ends with classification using Python Streamlit.

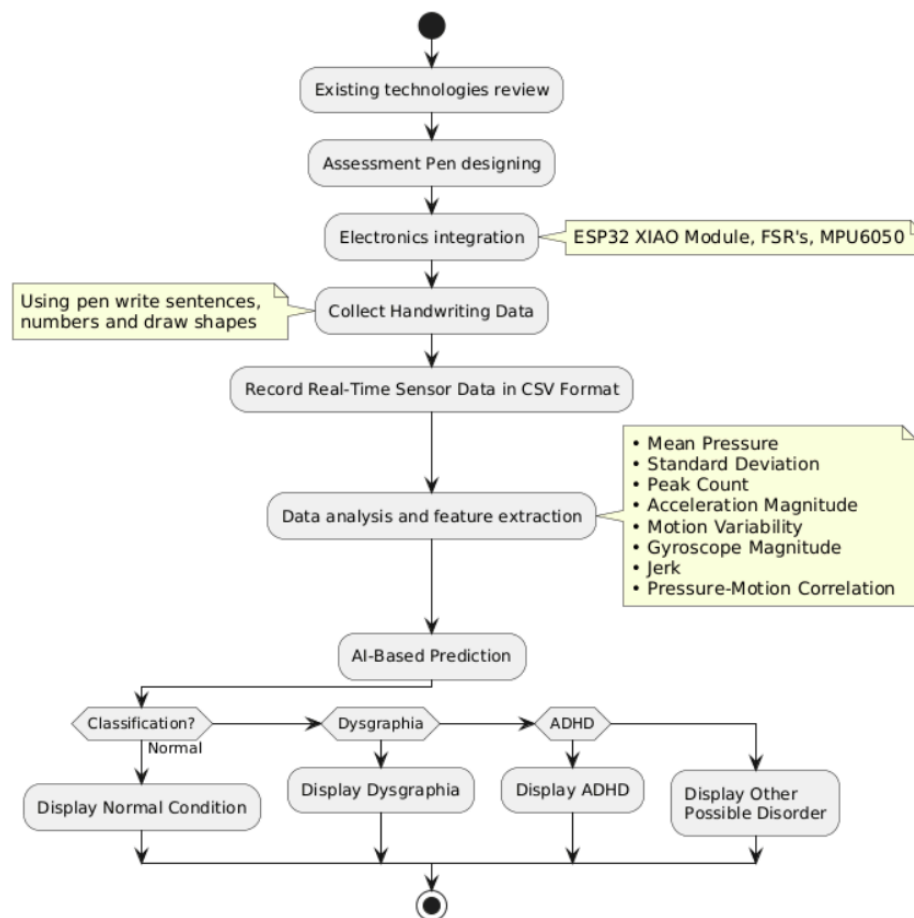


Figure 1: System flow diagram

Implementation Strategy: The implemented system has been used in a step-by-step research process which includes six key phases: literature review, hardware design, handwriting data acquisition, dataset preparation, development of distance-based screening algorithm and diagnostic visualization. Firstly, a comprehensive study of the literature has been performed with regard to the biomechanics of handwriting, dysgraphia, Attention deficit hyperactivity disorder (ADHD), motor coordination disorders and existing digital handwriting assessment tools. This study helped to discover the main biomechanical components of handwriting disorders, such as gripping pressure, pressure during writing, the height of the movement, the speed of writing, stability of writing strokes and movement variability. Based on the acquired information, a smart device for handwriting data acquisition was developed by integration of pressure sensors and a motion measuring unit in a compact writing device. The actual prototype is as shown in Figure 2.



Figure 2: Actual pen prototype

Mechanical design: The device is implemented as a simple pen-type tool similar to a standard pen. Two Force Sensitive Resistors (FSR) are integrated into the pen, namely a sensor at a grip for measuring the gripping force of the thumb and another sensor close to the beginning of a writing process for measuring the pressure while writing on the paper. An MPU6050 with three-axis accelerometer and a three-axis gyroscope was embedded inside the pen to collect dynamic processes regarding handwriting movements such as acceleration, angle of motion, stability of strokes, and writing tremors. Finally, all electronic components are placed into a light-weight 3D-printed case that is strong enough to provide the necessary stability and comfort at writing. The CAD design is as shown in Figure 3.

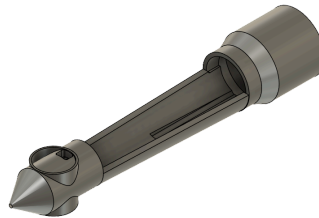


Figure 3: CAD Design of Pen

Electronics design: The electronic setup consists of the Seeed Studio XIAO ESP32-C3 Microcontroller, which acts as the main processing unit and communication interface for the system. There are two Force-Sensitive Resistors (FSRs) that are integrated into the system in order to measure pressure and one MPU6050 Inertial Measurement Unit (IMU) to track motion. The sensors will capture real-time handwriting dynamics that will then be processed and sent to a cloud-based dashboard through Wi-Fi. The circuit schematic is as shown in Figure 4.

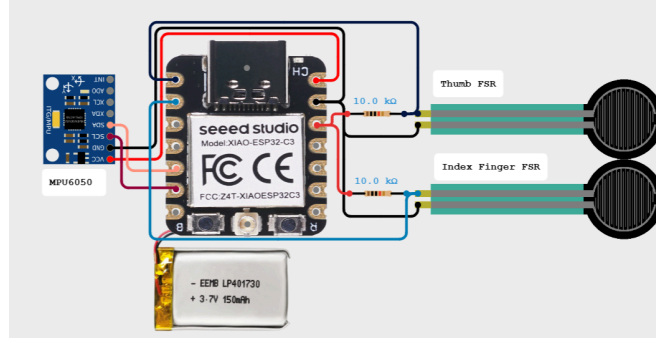


Figure 4: Circuit schematic of the smart pen

The circuit consists of two main sensing modules:

- (a) FSR Based pressure sensing - The FSRs are configured in voltage divider circuits with a fixed resistor (10 k Ω) as the second leg of the voltage divider. The FSR acts as a variable resistor, in which the resistor value decreases with increasing levels of applied force.

Working Principle

No pressure $\rightarrow$ High resistance $\rightarrow$ Low voltage output

High pressure $\rightarrow$ Low resistance $\rightarrow$ High voltage output

FSR Output Voltage is calculated as mentioned in Eqn (1).

$$V_{out} = V_{cc} \times \frac{R_{fixed}}{R_{FSR} + R_{fixed}} \quad (1)$$

where V_{out} = Force sensor output voltage

V_{cc} = Input voltage to FSR terminal = 3.3V

R_{fixed} = Voltage divider resistor = 10k Ω

R_{FSR} = Resistance due to pressure on sensor

Therefore it is stated that as the pressure on the sensor increases the FSR resistance will decrease and the output voltage will increase. The obtained voltage values are further converted into the ADC values between 0-4095. ADC Conversion is done using the formula in Eqn (2).

$$ADC = \frac{V_{out}}{V_{ref}} \times 4095 \quad (2)$$

Hence this is the amount of the pressure being applied on the different FSR sensors.

- (b) MPU6050 Motion Sensing circuit - The MPU6050 is connected via the I2C protocol, using two communication lines. It internally consists of a 3-axis accelerometer and 3-axis gyroscope. This enables real-time tracking of handwriting motion and stability.

Working Principle

Accelerometer measures linear motion (m/s²)

Gyroscope measures angular rotation (rad/s)

Acceleration Magnitude is given as A which is used to measure the total motion intensity independent of the direction in Eqn (3).

$$A = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad (3)$$

Where a_x = acceleration in X direction

a_y = acceleration in Y direction

a_z = acceleration in Z direction

Now the Gyroscope magnitude is given by G which indicates the rotational instability and the tremors are as shown in Eqn (4).

$$G = \sqrt{g_x^2 + g_y^2 + g_z^2} \quad (4)$$

where g_x = rotation about X axis

g_y = rotation about Y axis

g_z = rotation about Z axis

Hence the average for the mean given as μ for the average pressure and motion levels is given as shown in Eqn (5)

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

(5)

With the help of the mean one can determine the standard deviation given as σ which can be used to determine the possibility of dysgraphia and is shown in Eqn (6).

$$\sigma = \left(\sqrt{\frac{1}{N} \sum (x_i - \mu)^2} \right) \quad (6)$$

Now when the σ is determined to be low then it can be stated as a normal and stable writing condition whereas, if the σ is detected to be high then it is stated as the irregular motion and is an indication of dysgraphia.

Therefore meaningful behavior characteristics are extracted from the raw signals from the sensors through data processing. The pressure signals from the FSRs indicate the strength of the person's grip and the amount of pressure applied when they are writing, and the motion signals from the MPU6050 indicate the smoothness of the strokes made and the change in direction of the strokes. Increased variability in the motion signals indicates poor motor control, which is a defining feature of individuals with dysgraphia.

Handwriting Dataset Generation: After successful hardware evaluation, a smart pen was employed to attain handwriting data from test subjects and collect data under a range of writing conditions. Each participant was asked to conduct a series of predefined writing activities, which included: writing sentences, tracing the alphabet, free writing, and repeating writing tasks. During the course of the writing activities, the smart pen continuously logged grip pressure, writing pressure, linear acceleration, angular velocity, and temperature. The resulting sensor values were then synchronized and stored in CSV files, which would contain multiple timestamps for both pressure and motion measurements for each entry. Several handwriting samples were obtained from each of the subjects to show how the way of writing varies with different writing contexts. The resulting dataset is the basis for the next stage, which is feature extraction and analysis. The same is shown in Figure 5.

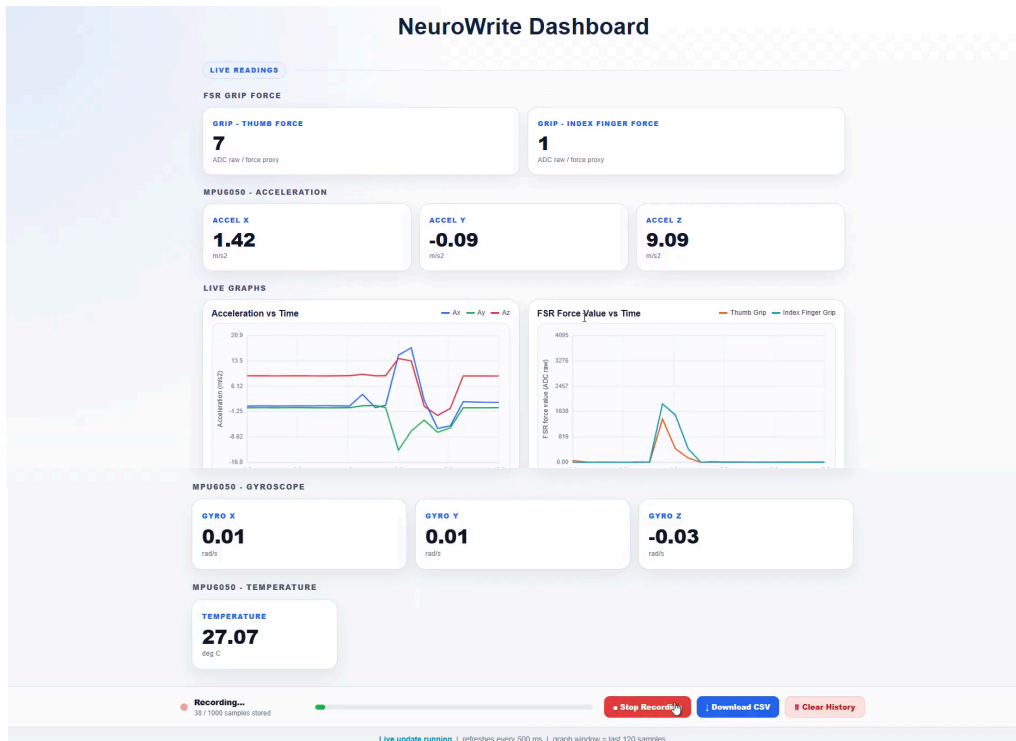


Figure 5: Data collection dashboard showing live sensor readings

Feature Extraction and Decision Support Dashboard: Brought into being by means of Streamlit, a Graphical User Interface (GUI) facilitates real-time analysis of handwriting sensor data obtained from the NeuroWrite smart pen. The platform enables users to upload a CSV file that is composed of sensor recordings gathered while performing writing tasks with Force Sensitive Resistor (FSR) and MPU6050 sensors. Once the file is uploaded, the system conducts feature extraction and measures the set of neurological biomarkers resulting from the writing activities. The features include - mean grip force, grip force variability, acceleration magnitude, motion variability, gyroscope value, jerk, and correlation between pressure and point movement. The features harvested are used to compare the collected information with an existing dataset that has readings from patients with ADHD, dysgraphia, and participants with normal handwriting development programs. To conduct the comparison, feature-space similarity measures are applied to associate the uploaded sample to the group with the most similar feature characteristics. This complete flow is mentioned in Figure 6. Finally confidence is computed from the normalized inverse distance between the uploaded feature vector and each reference group. The same has been shown in Figure 7.

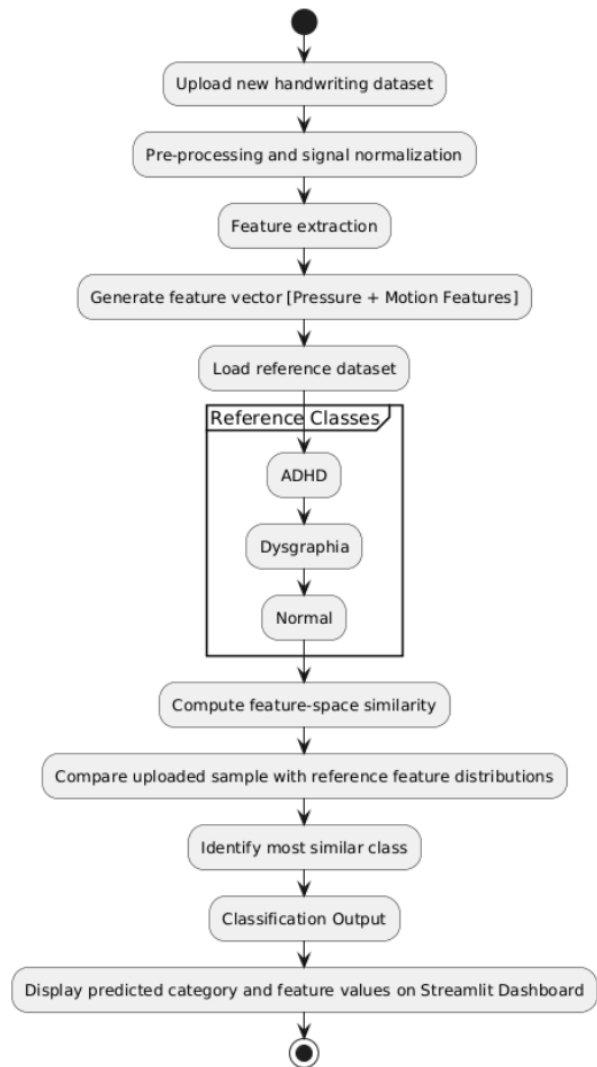


Figure 6: Execution workflow of the system

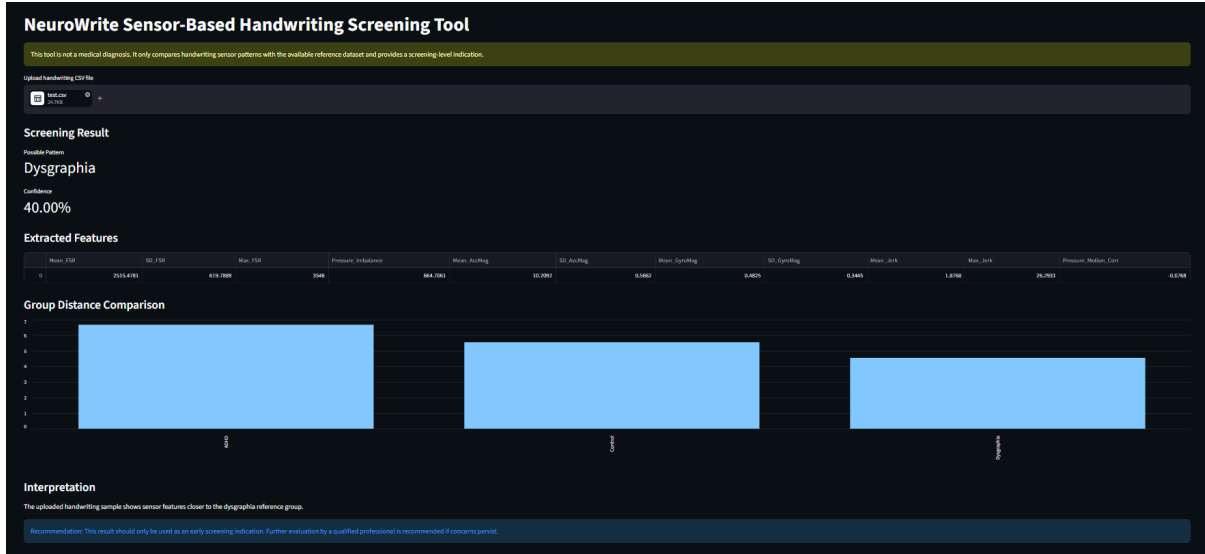


Figure 7: Data analysis dashboard

Results and Discussions

Smart pen performance: The NeuroWrite smart pen attained successful acquisition of handwriting sensor data from all participating subjects during writing and drawing events. The system employed two Force Sensitive Resistors (FSRs) located at the grip part of the pen and an MPU6050 inertial measurement unit (IMU) to enable motion detection. FSR sensors were constantly measuring grip pressure changes, while the MPU6050 was recording the acceleration and angular velocity of the subjects during handwriting. For all 60 participants (20 in Control, 20 in ADHD, and 20 in Dysgraphia groups), the system was completely successful in obtaining the required data, without any sensor failure or recording corruption throughout the procedures. This can be understood using Table 1.

Table 1: Smart Pen Data Acquisition Summary

Group	Participants	Successful Recordings	Data Loss Rate
Control	20	20	0%
ADHD	20	20	0%
Dysgraphia	20	20	0%
Total	60	60	0%

The recorded signals were processed to extract quantitative handwriting biomarkers as mentioned in Table 2. Statistical significance analysis will be conducted in future work due to the limited sample size. The average feature values across groups are summarised below.

Table 2: Mean Handwriting Biomarkers by Group

Feature	Control (Mean $\pm$ SD)	ADHD (Mean $\pm$ SD)	Dysgraphia (Mean $\pm$ SD)
Mean Grip Pressure	0.42 $\pm$ 0.08	0.51 $\pm$ 0.10	0.63 $\pm$ 0.12
Pressure Variability	0.18 $\pm$ 0.05	0.27 $\pm$ 0.07	0.35 $\pm$ 0.09
Motion Variability (m/s ²)	0.21 $\pm$ 0.06	0.38 $\pm$ 0.09	0.44 $\pm$ 0.11
Gyroscopic Instability (rad/s)	0.19 $\pm$ 0.04	0.33 $\pm$ 0.08	0.41 $\pm$ 0.10
Jerk (Movement Smoothness) (m/s ³)	0.17 $\pm$ 0.05	0.29 $\pm$ 0.07	0.39 $\pm$ 0.08
Pressure-Motion Correlation (Dimensionless)	0.72 $\pm$ 0.06	0.58 $\pm$ 0.09	0.46 $\pm$ 0.10

The outcomes have clearly differentiated the groups with dysgraphia exhibiting the maximum variability of motion parameters while the control subjects showed consistent patterns of writing. Few important findings are that -

1. Dysgraphic individuals displayed ~45% higher motion variability than controls.
2. The ADHD group showed limited ability to coordinate the effects of pressure and motion, which indicates inadequate consistency of motor control.
3. The control participants demonstrated the greatest stability of pressure and the lowest scores of jerk indicating smooth handwriting execution.

Such quantitative empirical data confirm the effectiveness of the smart pen in capturing specific neuromotor features of different writing profiles making it an efficient objective assessment tool.

Dashboard Analysis and Screening Results: A NeuroWrite dashboard has been created utilizing Streamlit and Python to create an automated handwriting screening system. Users can upload sensor data (in CSV format) of the smart pen to the NeuroWrite system. The system then undergoes processes such as preprocessing, feature extraction, and comparison with control, ADHD, and Dysgraphia data. The dashboard measures several neuromotor biomarkers such as grip force, pressure variability, motion variability, gyroscope instability, jerk, and coordination ratio. It will use its classification method based on similarity to classify the acquired data by comparing it to the data of the reference group. These features include mean grip pressure, pressure variability, maximum pressure, acceleration magnitude, gyroscope magnitude, jerk, and pressure-motion correlation. The extracted feature vector is normalized and compared with the reference feature vectors obtained from the Control, ADHD, and Dysgraphia datasets. Euclidean distance is calculated between the uploaded sample and each reference group. The group with the minimum overall distance is presented as the most probable screening result together with a confidence score.

The dashboard outputs for Control, ADHD, and Dysgraphia classifications are displayed in Figures 8, 9 and 10 respectively, showing clear distinctions between the handwriting patterns in different groups.

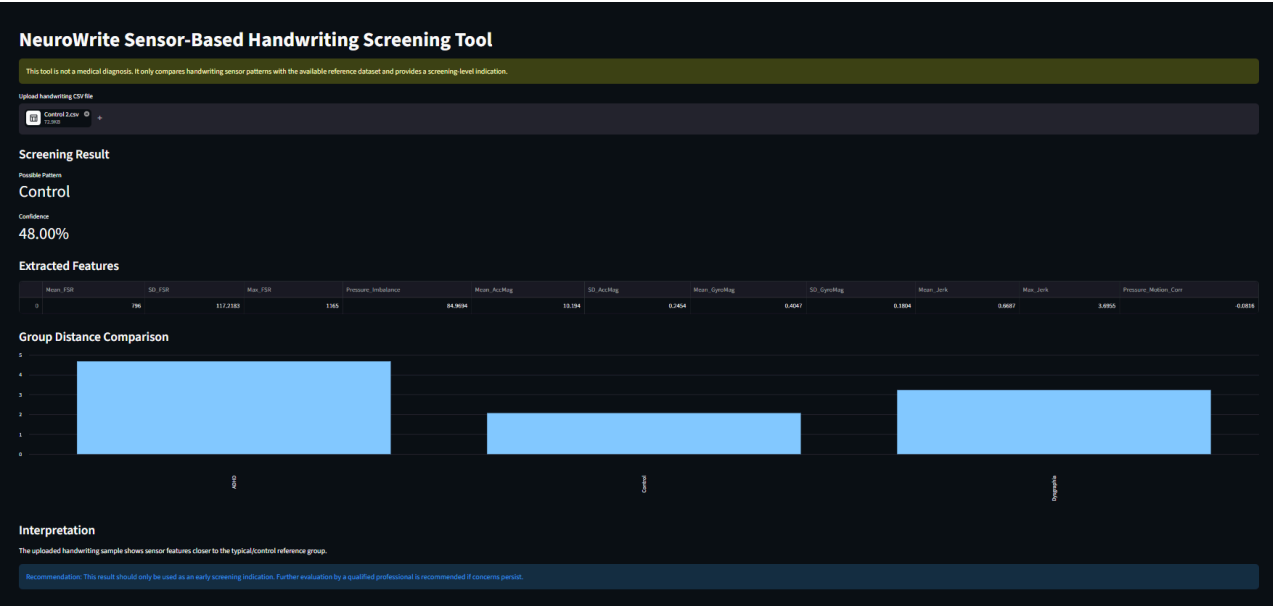


Figure 8: Control Sample classified correctly

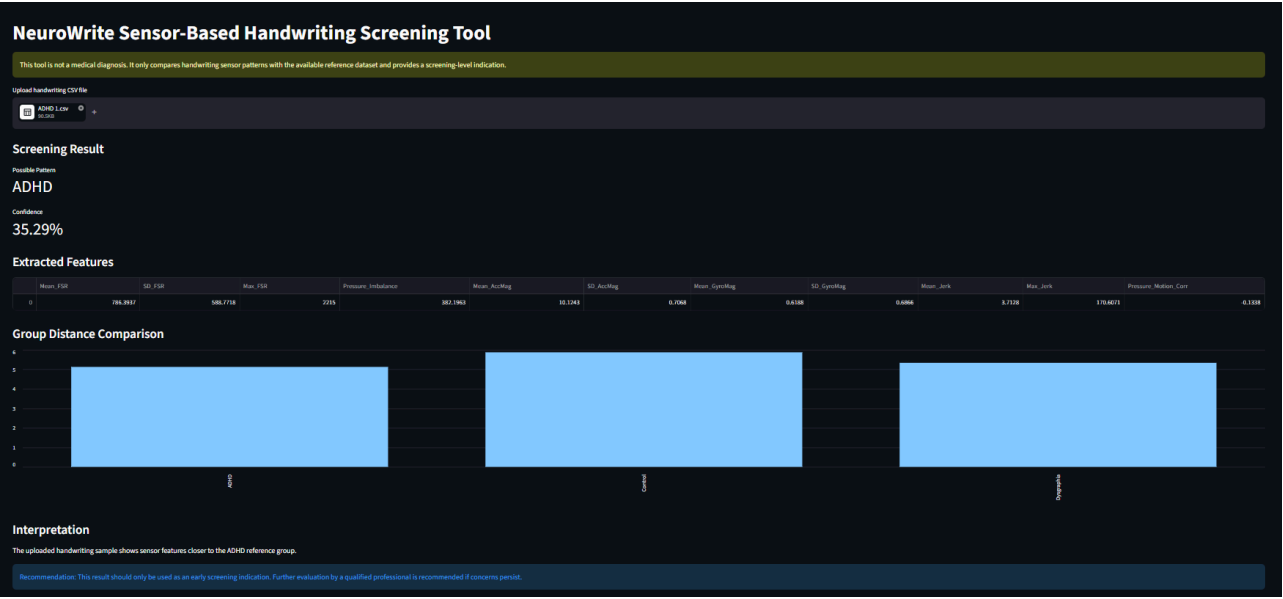


Figure 9: ADHD Sample showing increased pressure variability

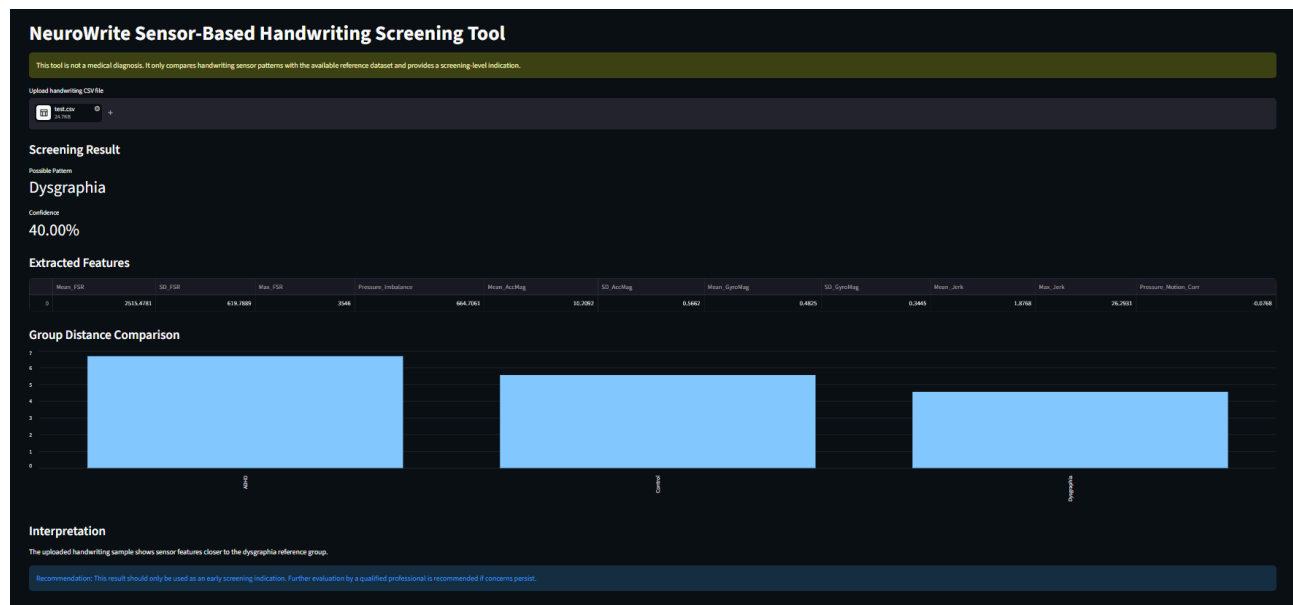


Figure 10: Dysgraphia Sample showing highest motion variability

The NeuroWrite system is aimed at being an early screening and decision-making system rather than being a diagnostic tool helping identify those individuals who need to undergo a more specialized assessment.

Conclusion

In contrast to conventional handwriting assessment techniques based on observational traits, the new technology consists of a smart pen that can detect how much pressure the user applies during handwriting. The study introduced NeuroWrite, a low-cost sensor-based smart pen created for the early detection of dysgraphia and handwriting problems caused by Attention deficit hyperactivity disorder (ADHD). The system works by using exactly two Force sensitive resistors (FSRs) and an MPU6050 inertial measurement processor to measure grip pressure and neuromotor performance during the process of writing. A sample that consists of 60 participants was collected and tested: 20 from the Control group, 20 with ADHD, and 20 with dysgraphia. The analysis yielded information relating to grip pressure, motion instability, gyroscopic instability, jerk, and correlation between pressure and motion. The outcomes showed the distinct differences between the tested groups. Dysgraphia group samples were marked by high levels of motion irregularity, while ADHD samples had higher levels of pressure instability. And most importantly, the NeuroWrite software allows for automated analysis and screening. The future research work shall be aimed at expanding the sample, including the other neuromotor disorders, using advanced machine learning methods, and testing the system on patients. It can further include CNN/LSTM based classification, real-time prediction without CSV upload along with proven clinical validation.

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